AN ORDER-THREE COUNTEREXAMPLE TO THE BARÁT–KORONDI–VARGA PROJECTION-AREA CONJECTURE

Abstract. Barát, Korondi, and Varga conjectured that every solution to the three-dimensional cube-dismantling problem has total orthogonal projection area at least $n^2 + 6n - 4$. We point out that the exhaustive order-three catalogue of Barát and Wanless already contains a counterexample. In coordinates, one such solution has projection sizes $6, 7, 7$, and hence total projection area $20 < 23$. We give a legal build-up certificate, and the same catalogue yields the exact value $s_3(3) = 20$. The configuration itself is not new; the contribution is the resulting correction to the stated conjecture.

Write $[n] = \{1, \dots, n\}$ and give $[n]^3$ its grid adjacency: two vertices are adjacent if they differ by 1 in exactly one coordinate. A *base position* is an independent set $B \subseteq [n]^3$ of size n^2. It is a *solution* if $[n]^3$ can be built from B by adding the vertices of $[n]^3 \setminus B$ one at a time, each added vertex having exactly three neighbours already present. This is the reverse of a legal cube dismantling.

For $i \in \{1, 2, 3\}$, let π_i delete the ith coordinate and set

$$A(B) = \sum_{i=1}^{3} |\pi_i(B)|, \qquad s_3(n) = \min\{A(B) : B \text{ is a solution in } [n]^3\}.$$

In the notation above, a conjecture of Barát, Korondi, and Varga [1], stated as Conjecture 1 of Barát and Wanless [2],[1] asserts

$$s_3(n) \geq n^2 + 6n - 4 \qquad \text{for all } n.$$

Theorem 1. *One has*

$$s_3(3) = 20 < 23 = 3^2 + 6 \cdot 3 - 4.$$

Consequently, the Barát–Korondi–Varga projection-area conjecture is false as stated.

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[1]The Barát–Korondi–Varga formulation is quoted here at second hand. Their extended abstract appeared in a symposium proceedings volume that is not publicly available: no copy is indexed by arXiv, zbMATH, Crossref, OpenAlex or Semantic Scholar, and the symposium's own abstract repository is restricted to participants. We have therefore not seen their statement, and in particular cannot confirm what number it carries there; [2] attribute it to [1] without one. The inequality displayed here, including the quantifier range "for all n", is verbatim the formulation printed as Conjecture 1 of [2], and that formulation is what the theorem below refutes.

Proof. For compactness, write ijk for (i,j,k). Consider

$$B = \{111, 113, 122, 131, 223, 311, 322, 331, 333\}.$$

No two vertices in B are adjacent, so B is a base position. Its projections are

$$\pi_1(B) = \{(1,1),(1,3),(2,2),(2,3),(3,1),(3,3)\},$$
$$\pi_2(B) = \{(1,1),(1,2),(1,3),(2,3),(3,1),(3,2),(3,3)\},$$
$$\pi_3(B) = \{(1,1),(1,2),(1,3),(2,2),(3,1),(3,2),(3,3)\}.$$

Thus $A(B) = 6 + 7 + 7 = 20$.

It remains to certify that B is a solution. Put $S_0 = B$, and for a set S let $N_S(v)$ be the set of neighbours of v in S. In the following table, set $S_k = S_{k-1} \cup \{v_k\}$. Each displayed set is the full current neighbourhood of the added vertex.

k	v_k	$N_{S_{k-1}}(v_k)$	k	v_k	$N_{S_{k-1}}(v_k)$
1	112	$\{111, 113, 122\}$	10	232	$\{222, 231, 332\}$
2	121	$\{111, 122, 131\}$	11	233	$\{223, 232, 333\}$
3	123	$\{113, 122, 223\}$	12	132	$\{122, 131, 232\}$
4	222	$\{122, 223, 322\}$	13	133	$\{123, 132, 233\}$
5	321	$\{311, 322, 331\}$	14	211	$\{111, 221, 311\}$
6	323	$\{223, 322, 333\}$	15	212	$\{112, 211, 222\}$
7	332	$\{322, 331, 333\}$	16	213	$\{113, 212, 223\}$
8	221	$\{121, 222, 321\}$	17	312	$\{212, 311, 322\}$
9	231	$\{131, 221, 331\}$	18	313	$\{213, 312, 323\}$

Every current neighbourhood has cardinality three, and $\{v_1, \ldots, v_{18}\} = [3]^3 \setminus B$. Hence the table is a legal build-up from B to $[3]^3$, proving $s_3(3) \le 20$.

For the reverse inequality, Barát and Wanless exhaustively enumerated all order-three solutions and found seven isometry classes, one representative of each being drawn in their Figure 6 [2, Section 4, Table 1 and Figure 6]. Computing the projection totals of those seven representatives gives

$$27,\ 27,\ 25,\ 23,\ 21,\ 20,\ 21,$$

of which only the multiset is used here. A cube isometry preserves total projection area, so these are the totals of the seven classes. Their classification therefore implies $s_3(3) \ge 20$, and hence $s_3(3) = 20$. The same conclusion is reached without the figure, by the exhaustive enumeration recorded at the end of this note: of the 4224 independent 9-subsets of $[3]^3$, exactly 116 are solutions, they fall into seven isometry classes carrying the same seven totals, and the least of those totals is 20. □

Prior work. The set B is isometric to one of the seven representatives in Figure 6 of [2], necessarily the one of total projection area 20, since exactly one of the seven classes attains that value. Thus the configuration, its validity as a solution, and the classification that pins $s_3(3)$ were all already contained in the published order-three catalogue; the observation recorded here is that the resulting value contradicts the conjectured bound.

Status and scope. What is refuted is the formulation displayed above, namely the one printed as Conjecture 1 of [2]; on the sourcing of that wording, and on what can and cannot be checked about the numbering it carries in [1], see the footnote there. The failure is not confined to the minimiser: three of the seven order-three isometry classes have total projection area below 23, namely 20, 21 and 21. Nor is the bound false at every order. It holds with equality at the two smaller ones, $s_3(1) = 3$ and $s_3(2) = 12$, so $n = 3$ is the first order at which it fails. No claim is made about $s_3(n)$ for $n \geq 4$.

Exact verification. The computation reported above was re-run independently, on a separate machine, by a standalone program that takes the data printed here as input and recomputes everything else from the definitions in exact integer arithmetic. All 32 of its checks agree with what is stated here. It rebuilds the grid graph on $[3]^3$ and the addition rule, and separates the rule "exactly three neighbours already present" from "at least three" by exhibiting configurations on which the two readings differ; it re-derives the independence of B, the three projection sets elementwise, and $A(B) = 6 + 7 + 7 = 20$; it replays all eighteen rows of the table, requiring each recomputed neighbourhood to equal the printed one and to have exactly three elements, and it finds a legal addition order of its own without consulting the table; and it carries out the order-three census afresh, enumerating all 4224 independent 9-subsets of $[3]^3$, testing each for solution-hood, building and verifying the 48-element cube isometry group, checking that A is constant on isometry classes, and obtaining 116 solutions in seven classes with least total 20. The count 4224 is confirmed by a second, unrelated algorithm, a layer transfer-matrix dynamic program; the same census at $n = 1$ and $n = 2$ returns equality in the conjectured bound, a control against systematic undershoot; and a mutation test reports failure under each of a suite of deliberate corruptions of the printed data and of the program's own geometry, with one exception that is provably immaterial on this input, namely interchanging the first two rows of the table, whose printed neighbourhoods both lie inside B. No file accompanies this note: every quantity above is recomputable from the data printed here together with the definitions of the first two paragraphs, and the program used and its transcript are retained in an auxiliary archive, available on request.

Two things the re-run did not reproduce. First, nothing beyond $n = 3$: a census at $n = 4$ would have to test 15,183,071,352 independent 16-subsets of $[4]^3$, about 3.6 million times the order-three workload, and it was not attempted—nor is anything beyond $n = 3$ claimed here. Second, the attributions, which are assertions about the literature rather than about mathematics: that the inequality transcribed above is the one posed in [1], and that the seven representatives drawn in Figure 6 of [2] are those of the seven isometry classes whose totals are computed above—only the multiset of those totals is used, which is all the theorem needs. Those are readings of the cited papers, and in the case of [1] of a source we were unable to consult at all. The re-run agrees with every number printed here; agreement is evidence that the computation is right, not a proof of it.

References

[1] J. Barát, M. Korondi, and V. Varga, *How to dismantle an atomic cube in zero gravity*, in *Proceedings of the 7th Hungarian–Japanese Symposium on Discrete Mathematics and Its Applications*, Kyoto, Japan, 2011; extended abstract, not publicly available. The details of this entry are those printed in the reference list of [2]; we have not seen the volume.

[2] J. Barát and I. M. Wanless, *A cube dismantling problem related to bootstrap percolation*, J. Stat. Phys. **149** (2012), 754–770, doi:10.1007/s10955-012-0622-7; arXiv:2603.23913 [math.CO].